INSTRUCTIONS

GEK-49759A

Supersedes GEK-49759

Metal-Clad Switchgear

Documentation

GENERAL ELECTRIC

NOTES

METAL-CLAD SWITCHGEAR DOCUMENTATION

INTRODUCTION

These instructions are provided as a practical guide in the use of the various documents contained in the documentation package furnished with General Electric Company metal-clad switchgear. They will prove useful to those who install, operate or maintain the switchgear.

The user will find that there are three separate but interrelated types of documents. They are:

Equipment Summary
Front View and Floor Plan
Electrical Diagrams

EQUIPMENT SUMMARY

The equipment summary (bill of material) is identified by a unique summary number. This document lists all the circuit breakers, relays, instruments, meters, control and transfer switches and other similar devices that comprise the equipment package. Each item of the summary is identified by a mark number which may consist of three numbers, three letters or some combination of letters and numbers - for example; 071, OBK, 10S, 20P are all mark numbers.

Devices shown on the front view drawing are identified with mark numbers. Devices not shown on the front view are mounted internally in the equipment, or on side sheets or swinging panels. These devices are identified by mark numbers on the connection diagrams.

Certain "stock" items furnished as part of the equipment are not included in the summary. Items in this category are terminal boards, relay blocks, fuse blocks and other similar components.

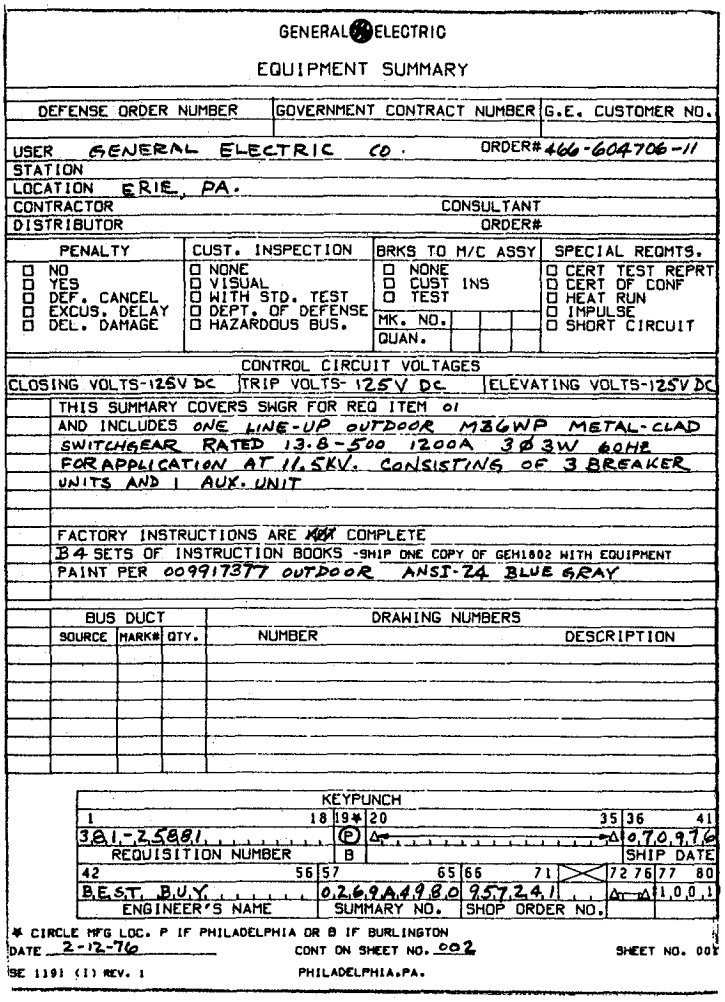
The equipment summary is made up of several main parts.

1. Front Sheet

   It contains a brief description of the equipment supplied. It identifies the user, contractor, consultant and agent as well as their applicable purchase order numbers. In addition, it identifies (1) General Electric Company requisition number (2) shop order number and (3) assigned requisition engineer. Items 1, 2 and 3 are repeated on all summary sheets. A typical front sheet is shown in Figure 1.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

To the extent required the products described herein meet applicable ANSI, IEEE and NEMA standards; but no such assurance is given with respect to local codes and ordinances because they vary greatly.



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EQUIPMENT SUMMARY

DEFENSE ORDER NUMBER	GOVERNMENT CONTRACT NUMBER	G.E. CUSTOMER NO.

USER GENERAL ELECTRIC CO. ORDER# 466-604706-11
STATION
LOCATION ERIE, PA.
CONTRACTOR CONSULTANT
DISTRIBUTOR ORDER#

PENALTY	CUST. INSPECTION	BRKS TO M/C ASSY	SPECIAL REQMTS.
☐ NO ☐ YES ☐ DEF. CANCEL ☐ EXCUS. DELAY ☐ DEL. DAMAGE	☐ NONE ☐ VISUAL ☐ WITH STD. TEST ☐ DEPT. OF DEFENSE ☐ HAZARDOUS BUS.	☐ NONE ☐ CUST INS ☐ TEST MK. NO. QUAN.	☐ CERT TEST REPRT ☐ CERT OF CONF ☐ HEAT RUN ☐ IMPULSE ☐ SHORT CIRCUIT

CONTROL CIRCUIT VOLTAGES

CLOSING VOLTS-125V DC | TRIP VOLTS- 125V DC | ELEVATING VOLTS-125V DC

THIS SUMMARY COVERS SWGR FOR REQ ITEM 01
AND INCLUDES ONE LINE-UP OUTDOOR M36WP METAL-CLAD SWITCHGEAR RATED 13.8-500 1200A 3Ø 3W 60HZ FOR APPLICATION AT 11.5KV. CONSISTING OF 3 BREAKER UNITS AND 1 AUX. UNIT

FACTORY INSTRUCTIONS ARE NOT COMPLETE
34 SETS OF INSTRUCTION BOOKS -SHIP ONE COPY OF GEH1802 WITH EQUIPMENT
PAINT PER 009917377 OUTDOOR ANSI-74 BLUE GRAY

BUS DUCT			DRAWING NUMBERS	
SOURCE	MARK#	QTY.	NUMBER	DESCRIPTION

KEYPUNCH

1	18	19*	20	35	36	41
381-25881		P	Δ	Δ	070976	
REQUISITION NUMBER		B			SHIP DATE	

42	56	57	65	66	71	72	76	77	80
B.E.S.T. B.U.Y.		02694A4980		957241		Δ	Δ	1001	
ENGINEER'S NAME		SUMMARY NO.		SHOP ORDER NO.					

* CIRCLE MFG LOC. P IF PHILADELPHIA OR B IF BURLINGTON
DATE 2-12-76 CONT ON SHEET NO. 002 SHEET NO. 001
SE 1191 (1) REV. 1 PHILADELPHIA,PA.

Figure 1 - Typical front sheet of summary.

2. Drawing List Sheet

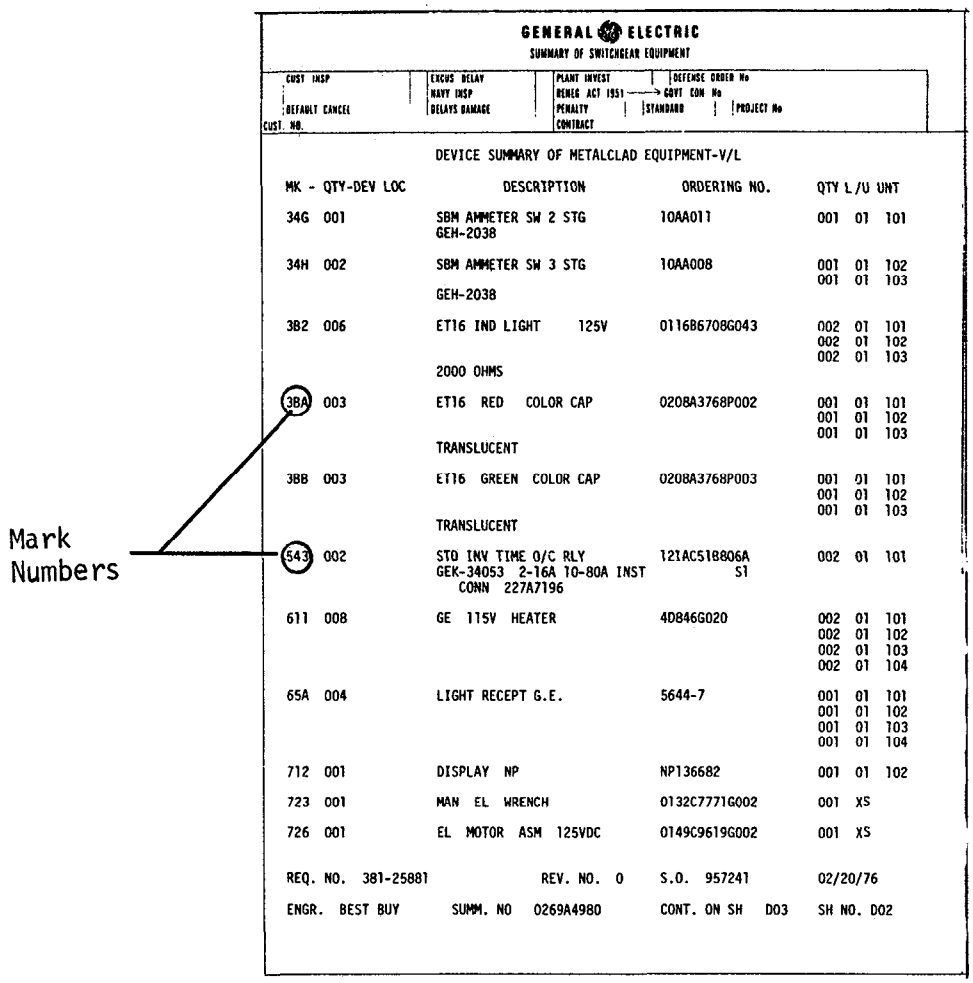
It lists drawing number and revision of all drawings applicable to the supplied equipment.

3. Breaker Sheets

Power circuit breaker description, quantity and mark number are listed.

4. Device Sheets - Computer Generated

These sheets identify devices by catalog number and quantity required. Their location in the equipment is identified by use of mark number, lineup number and unit number. A typical computer generated device sheet is shown in Figure 2.



GENERAL ELECTRIC

SUMMARY OF SWITCHGEAR EQUIPMENT

CUST INSP	EXCUS DELAY	PLANT INVEST	DEFENSE ORDER No
DEFAULT CANCEL	NAVY INSP	RENEG ACT 1951 →	GOVT CON No
CUST. NO.	DELAYS DAMAGE	PENALTY / STANDARD	PROJECT No
		CONTRACT	

DEVICE SUMMARY OF METALCLAD EQUIPMENT-V/L

MK - QTY-DEV LOC	DESCRIPTION	ORDERING NO.	QTY L/U UNT
34G 001	SBM AMMETER SW 2 STG GEH-2038	10AA011	001 01 101
34H 002	SBM AMMETER SW 3 STG GEH-2038	10AA008	001 01 102 001 01 103
3B2 006	ET16 IND LIGHT 125V 2000 OHMS	0116B6708G043	002 01 101 002 01 102 002 01 103
3BA 003	ET16 RED COLOR CAP TRANSLUCENT	0208A3768P002	001 01 101 001 01 102 001 01 103
3BB 003	ET16 GREEN COLOR CAP TRANSLUCENT	0208A3768P003	001 01 101 001 01 102 001 01 103
543 002	STD INV TIME O/C RLY GEK-34053 2-16A 10-80A INST CONN 227A7196	121AC51B806A S1	002 01 101
611 008	GE 115V HEATER	4D846G020	002 01 101 002 01 102 002 01 103 002 01 104
65A 004	LIGHT RECEPT G.E.	5644-7	001 01 101 001 01 102 001 01 103 001 01 104
712 001	DISPLAY NP	NP136682	001 01 102
723 001	MAN EL WRENCH	0132C7771G002	001 XS
726 001	EL MOTOR ASM 125VDC	0149C9619G002	001 XS

REQ. NO. 381-25881 REV. NO. 0 S.O. 957241 02/20/76

ENGR. BEST BUY SUMM. NO 0269A4980 CONT. ON SH D03 SH NO. D02

Mark Numbers

Figure 2 - Typical computer generated device sheet.

5. Device Sheet - Manually Generated

These sheets identify any required nameplate engraving and key interlocks by mark number.

FRONT VIEW AND FLOOR PLAN

A typical front view and floor plan drawing showing the physical arrangement of the equipment as well as the power circuit one line diagram is shown in Figures 3A and 3B.

An examination of Figures 3A and 3B reveals several points of interest. Some of these points are:

1. Front View

a. Instrument door mounted devices are shown. Included is the mark

number and some identification such as AM for ammeter or IAC for overcurrent relay.

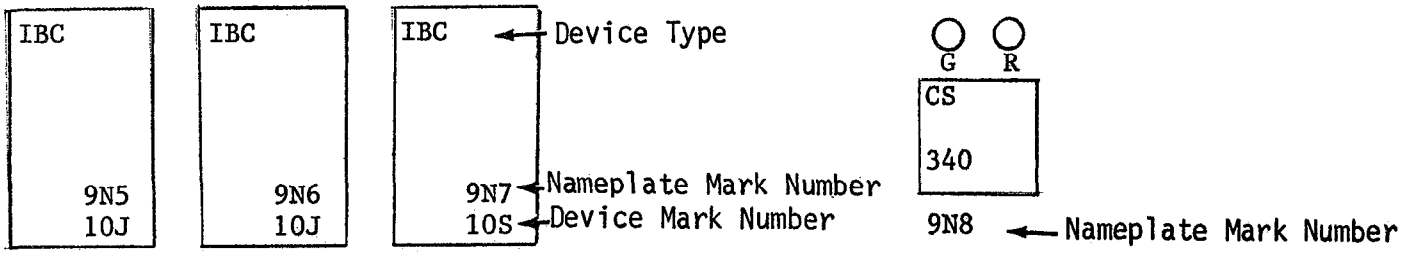
b. Each unit is individually dimensioned for width and is identified starting at the left end of the lineup by a G-E unit number. These unit numbers will always be in the hundreds series; i.e., 101, 102, 411, 604, etc.

c. The phase arrangement of the main bus is shown in the extreme left hand unit.

d. The shipping package dimensioning starts at the left hand end and progresses to the right. All vertical dimensions start at the mounting surface and progress to the top of the equipment. Dimensions are in decimal inches unless otherwise noted.

e. Weight of the total equipment including all devices and power circuit breakers is included.

f. When nameplates are required they will be assembled above the associated device with the nameplate mark number shown in accordance with the exhibit below.



2. Floor Plan

The floor plan is broken into two specific but interrelated parts. These are the general arrangement and the key view.

a. The general arrangement shows space for primary and secondary leads, the number of anchor bolt locations and the overall depth of the equipment.

b. The key view shows all dimensions for primary and secondary leads space, anchor bolt locations, door swing, recommended aisle spaces, etc.

Floor plan dimensions are progressive starting at the breaker removal end (front) and continuing to the rear of the unit.

In the case of outdoor equipment with a protected or common aisle arrangement, the aisle and the switchgear are shipped separately for field assembly. The weight of the aisle is shown. This weight is stated separately and is not a part of the total equipment weight referred to in 1.e. above. The G-E unit number is included on each unit of the aisle as an aid to field assembly. A multi-sheet front view and floor plan is supplied with this type of unit.

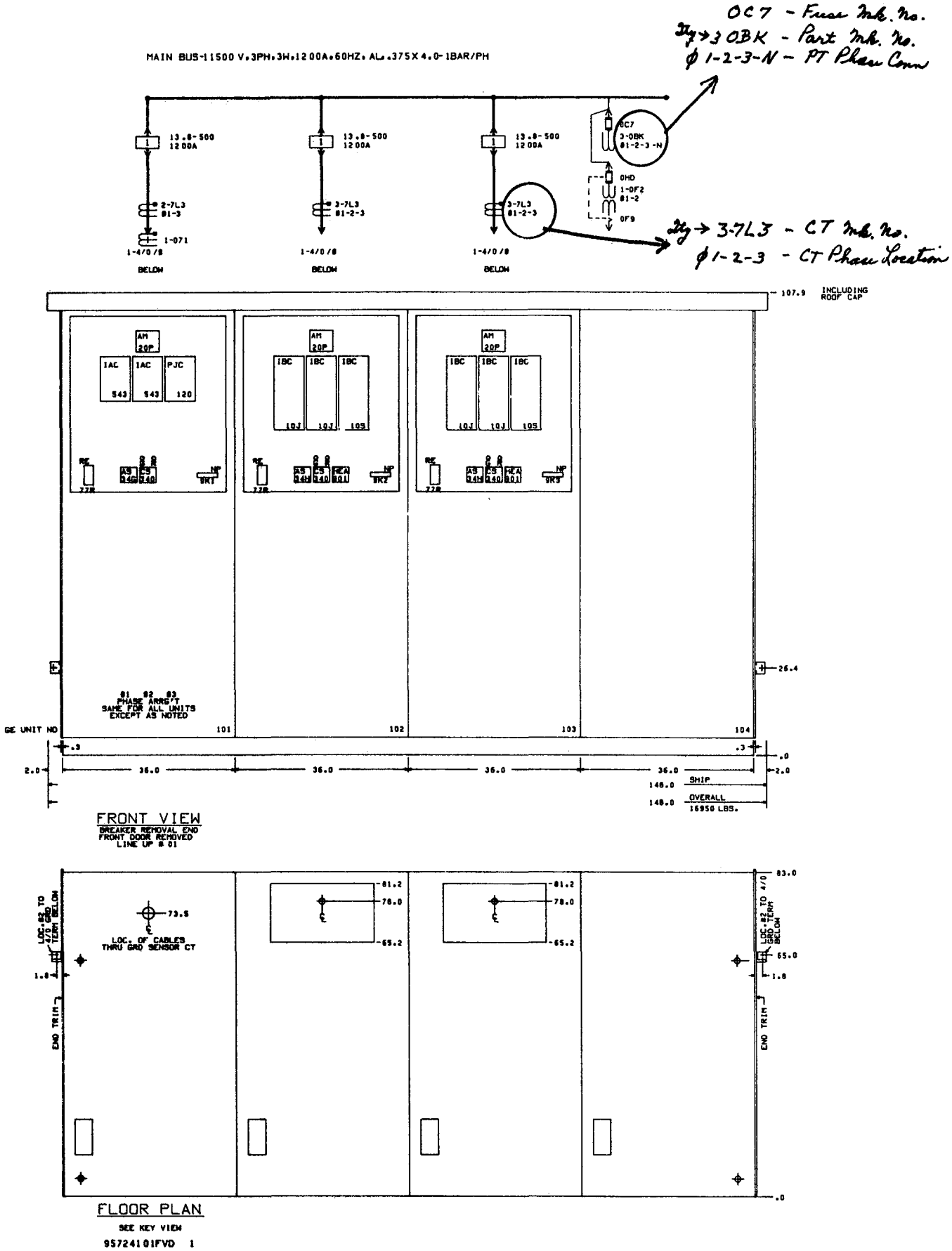


Figure 3A - Typical front view and floor plan
(See Figure 3B for key view)

3. One Line Diagram

The power circuit one line diagram is always shown and is arranged on a unitized basis. Associated with the one line diagram are such items as:

a. Power circuit breaker ratings and mark numbers.

b. Current transformers, their mark numbers and their phase locations.

c. Primary cable size and direction.

d. Potential and control power transformers, their fuses, their mark numbers and their phase connections.

e. Lightning arresters and their mark numbers.

f. Main bus rating along with its physical size and material.

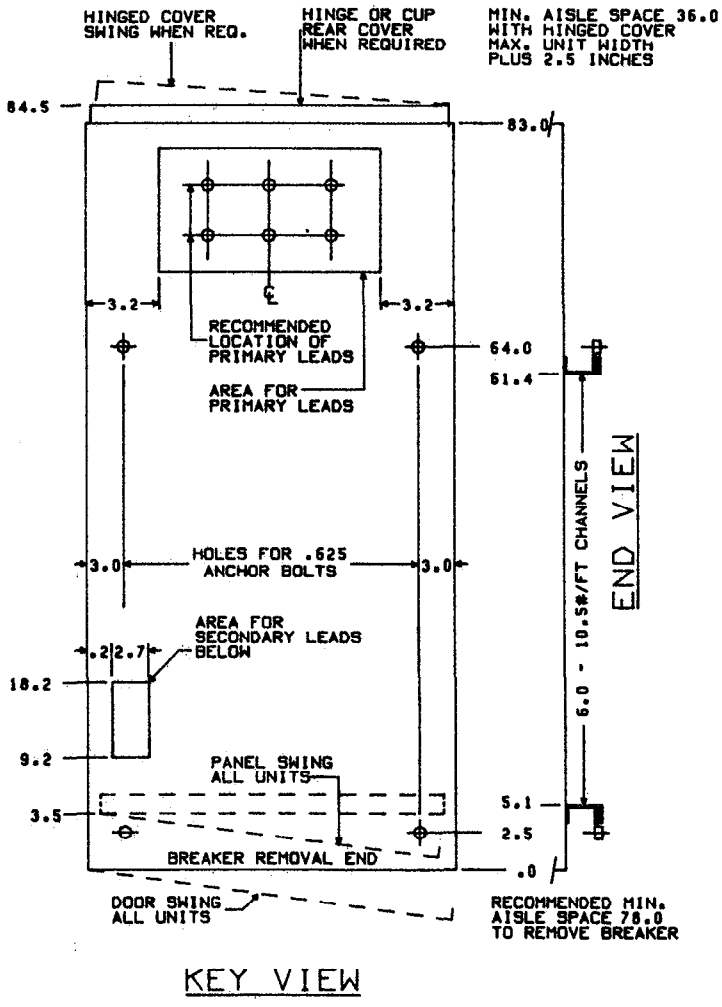


Figure 3B Part of front view and floor plan

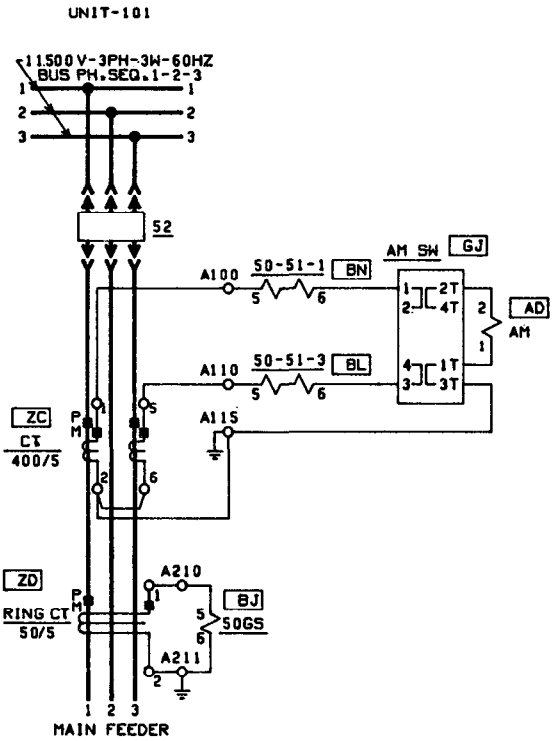
ELECTRICAL DIAGRAMS

The electrical diagram package consists of three main parts. The parts are:

Elementary Diagram
Connection Diagram
Interconnection Diagram

Elementary Diagram

The elementary diagram shows the circuitry as it exists in the delivered equipment. This detailed point-to-point representation allows the user to quickly identify and physically locate each wire. For each type of unit, the power circuit and associated control circuits are shown along with the actual termination of each wire either at a device or terminal board. Every device is functionally identified either by an American Standard Device Function Number (see page 19) or by abbreviations, such as AM for ammeter (see page 18). In addition, every device is identified by particular device identification letters appearing on or adjacent to each device. The device identification letters indicate the physical location of the device in the equipment.



All terminal points are shown on the elementary diagram. These include device terminal numbers, shipping split terminal boards, interconnection terminal boards and customer connection terminal boards. Control and transfer switch contact development and escutcheon positions are shown. For internal wiring of relays the user should refer to applicable relay instruction books supplied for the equipment.

Figure 4 illustrates part of a typical power circuit in three line form. Pertinent informational and/or instructional items are included. Figure 5 illustrates part of a typical control circuit on which are identified conventions used in its preparation.

Figure 4 - Typical elementary - power circuit

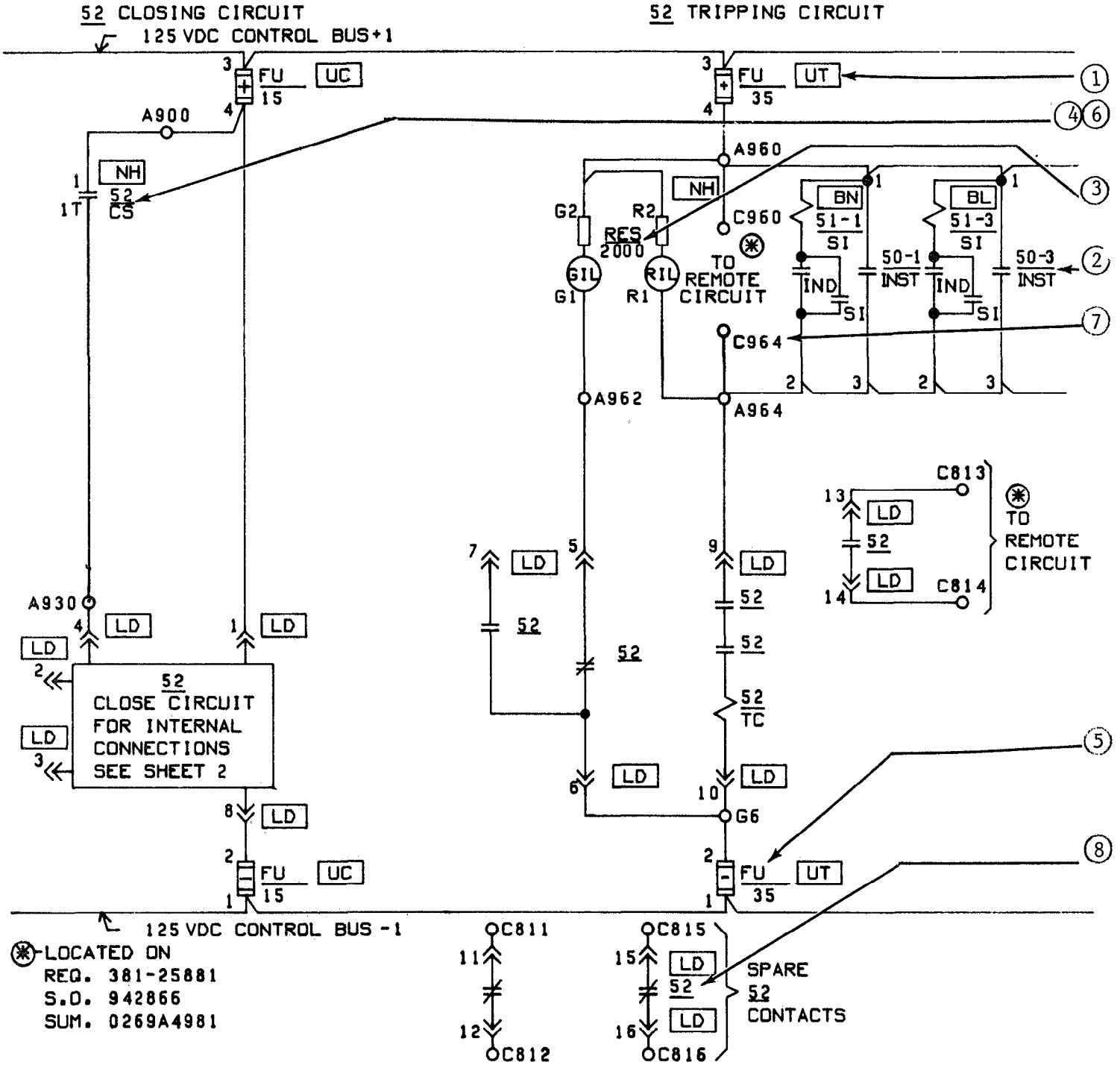


Figure 5 - Typical elementary - control circuit

The following conventions are used in the preparation of the elementary diagram. The numbered items are referenced on Figure 5 and illustrate the conventions used.

1. Device identification letters are shown in enclosed blocks near the device function number or abbreviation. ex. UT

2. Whenever applicable, devices are designated by American Standard Device Function Numbers. ex. 50

3. Devices to which function numbers are not applicable are given abbreviations. ex. RES

4. Distinguishing features or device elements such as rating and/or functions are frequently placed beneath the function number.
ex. 52/CS

5. The ampere ratings of fuses, total ohmic resistance of resistors, total micro-farad capacitance of capacitors are placed below the abbreviations.
ex. FU/35

6. Device terminal numbers are usually placed at the left of or below the graphic symbol.
ex. 1 / 1T

7. Shipping split, customer connection, instrument door and interconnection terminal board points are shown on the elementary diagram as small circles in the lines representing the wires.

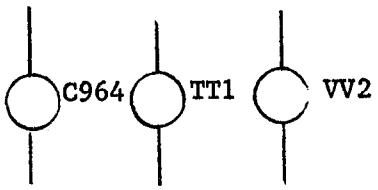
   Terminal boards are designated by a system of letters. The assigned letters depend on the function of the terminal board. Some of the designations and their functions are:

   a. Customer connection terminal boards
      C - standard - 30 Amp.
      TT, VV, WW, YY, ZZ - heavy duty - 100 Amp.

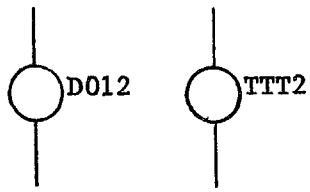
   b. Shipping split terminal boards
      D - standard - 30 Amp.
      TTT, VVV, WWW, YYY, ZZZ - heavy duty - 100 Amp.

   The terminal board points are labeled with significant circuit terminal designations. Each identification is a unique number in a computer system designation for a particular circuit. For instance, current circuits for metering and relaying are designated by the number series 100 thru 399.

ex.



Customer connection terminals

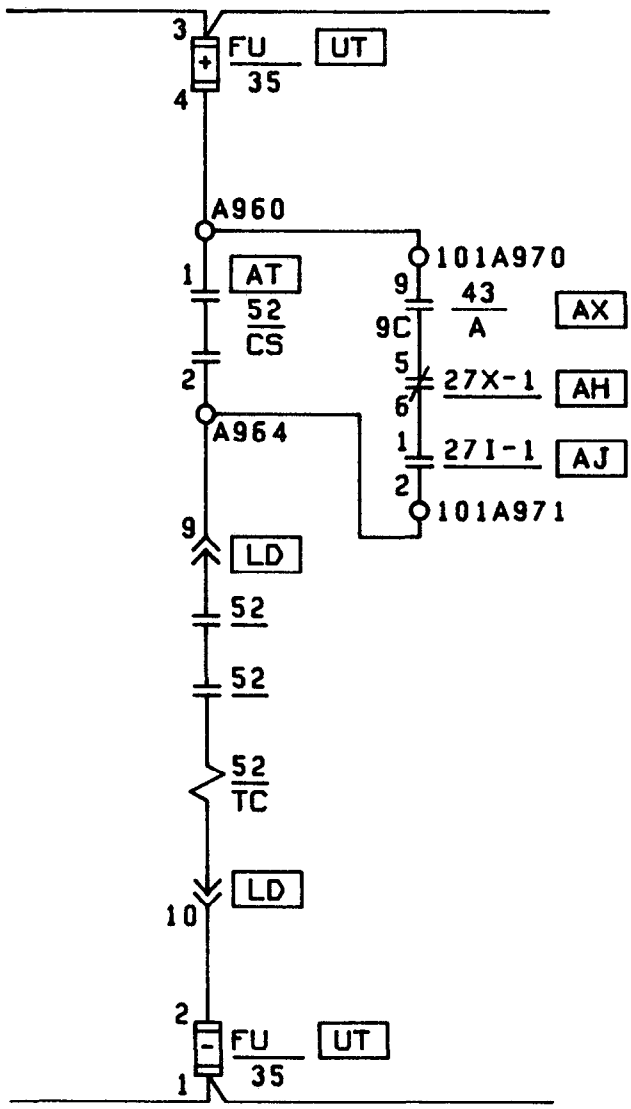


Shipping split terminals

8. The contacts of all devices are shown in their de-energized or non-operated positions, including those that operate in response to other than electrical quantities, such as temperature, speed, pressure, flow level, vibration, etc. The contacts of these devices which have no de-energized or non-operated position are shown normally open. Circuit breaker auxiliary contacts are shown in the position they assume when the circuit breaker main contacts are open. Sufficient information to indicate when the contacts are open or closed is shown.

ex  23 closes at 40F decreasing

9. Elementary diagram circuits are prepared for each type of unit. Devices physically located in other units of the line-up, but wired into the circuits of the unit under consideration, are identified through the point-to-point wiring system. Identification of end terminal points will be the method to clearly indicate that devices are mounted in another unit. In the example, terminal points A960 and A964 are located in unit 102. Terminal points 101A970 and 101A971 and all devices wired between them are located in unit 101 as indicated by the prefix 101.



ex. 52 tripping circuit - unit 102

Connection Diagram

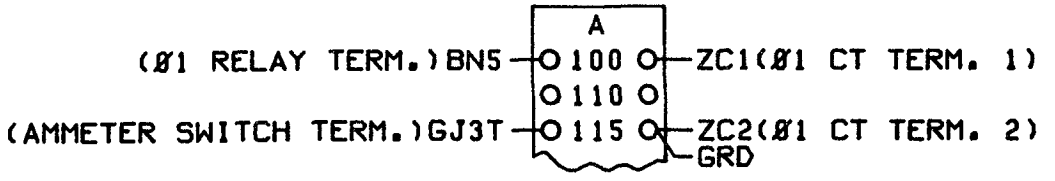
A separate connection diagram is prepared for each type of unit in a lineup of metal-clad switchgear. It consists of two main parts. Figure 6 illustrates a complete connection diagram.

Part of the diagram (see page 14) shows the general physical arrangement of all devices and accessory items such as terminal boards, fuse blocks, resistors, capacitors, etc. Adjacent to each device or accessory item is its device identification letter.

Inspection of Figure 6 reveals that the physical arrangement portion (page 14) and the wire list portion (page 15) of the connection diagram are each broken down into distinct parts. The distinct parts are determined by manufacturing techniques.

Certain conventions are used in the document preparation.

A. Figure 6, for example, has only one terminal board designated A (page 14). However, the wire lists (page 15) have three columns designated A. These three separate groups simply mean a wire originating at terminal board A has its opposite end terminating at some device located in a different wiring harness.



B. All control wires are AWG size 14 as standard. Any wire sizes different from the standard size 14 will have the AWG size designated. Exceptions to the use of size 14 for general wiring will be noted in the summary.

C. When the listed wire is followed by a letter it designates insulation covering or color.

ex. A = armored, G = green, W = white

D. Other wiring codes used are:

(M-J)	- metal jumper
(SPAR)	- spare terminal board point
(>)	- indicates more than one wire per terminal
(BUS + 1(#8))	- positive control bus number 1 and AWG size 8 wire
(GRD)	- ground to equipment frame
(G-B)	- ground bus connection
(LB)	- breaker elevating motor
(LC)	- breaker elevating motor clutch switch
(LE,LF)	- elevating motor limit switches

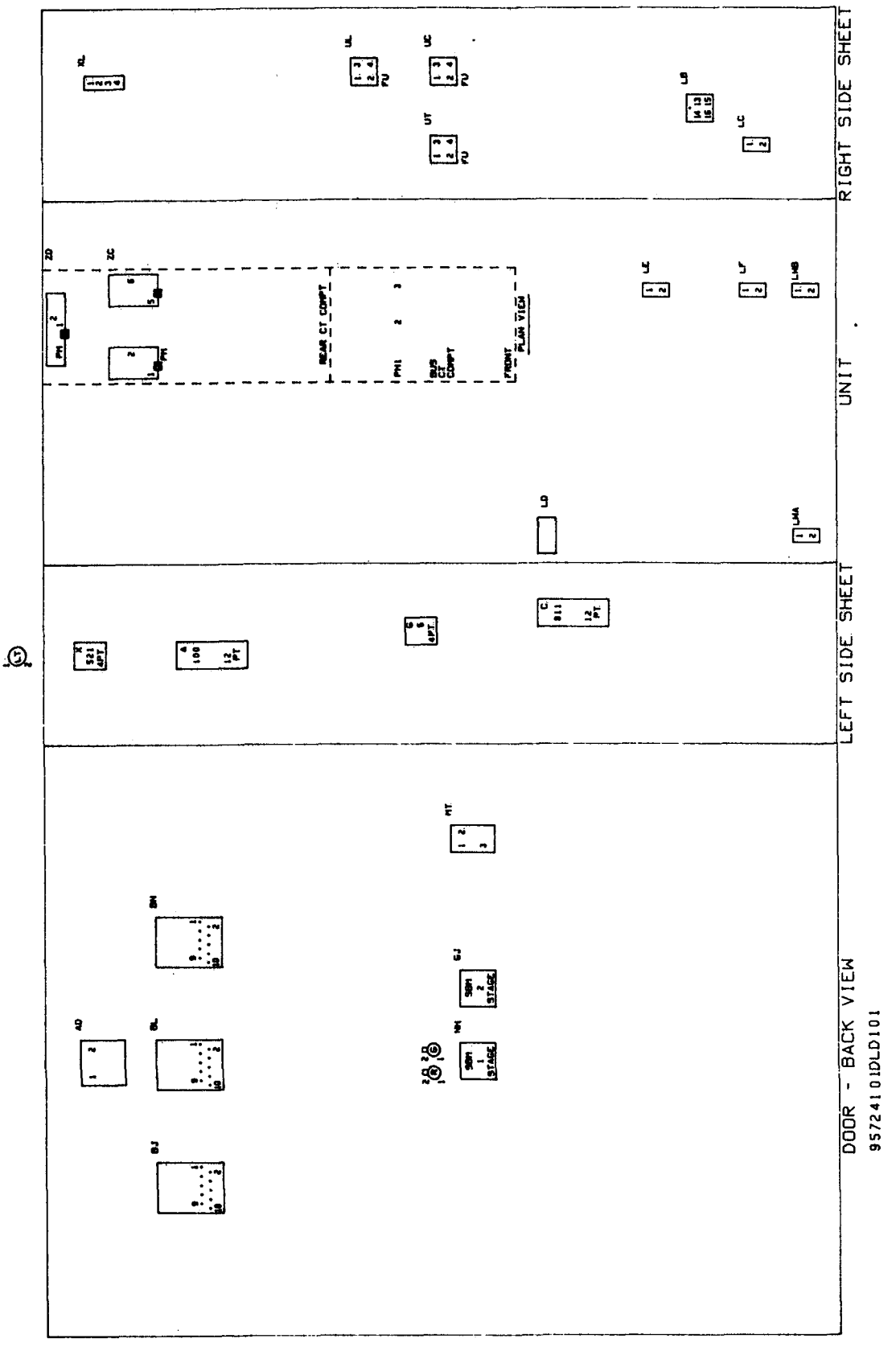
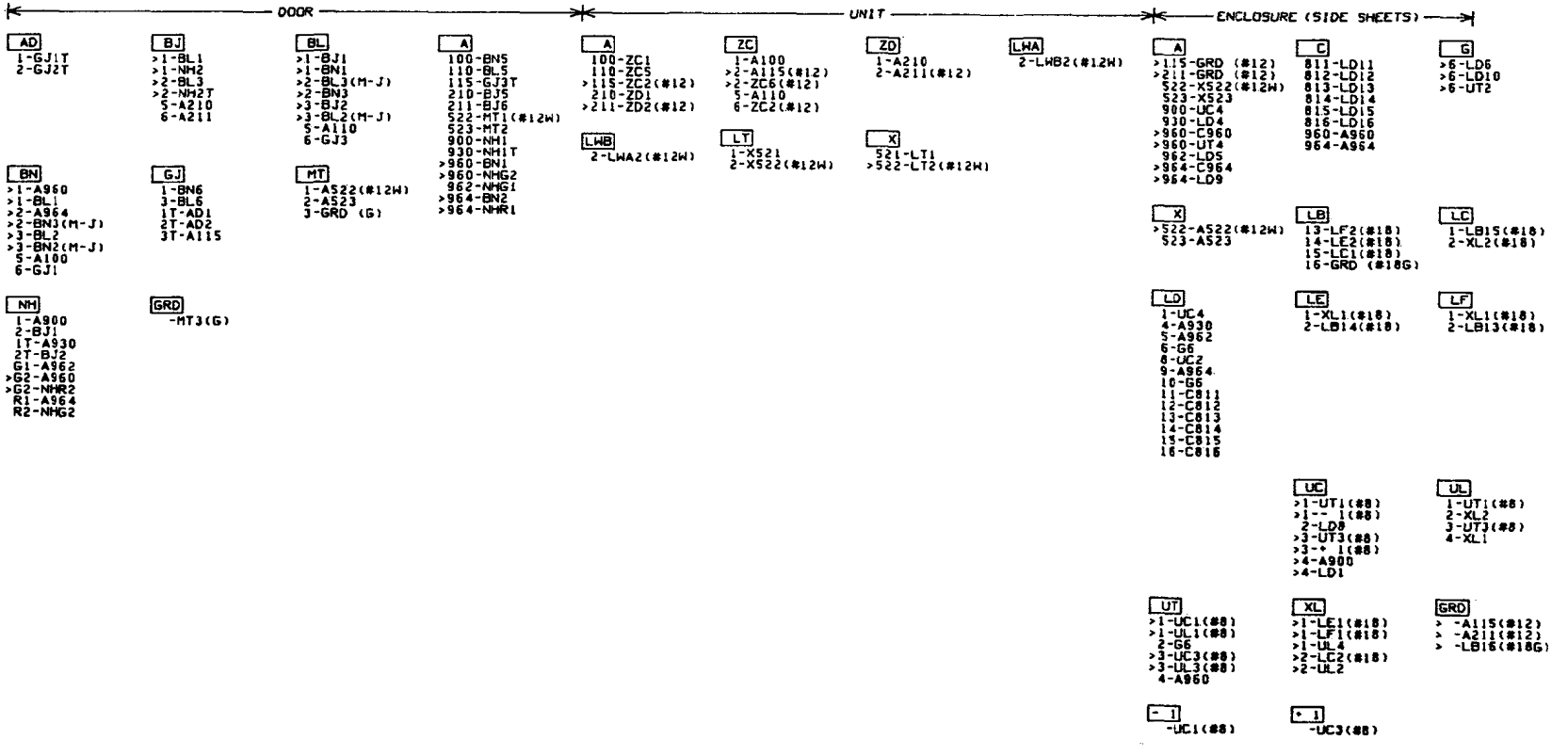


Figure 6 - Typical


DOOR
UNIT
ENCLOSURE (SIDE SHEETS)
AD
1-GJ1T
2-GJ2T
BJ
>1-BL1
>1-NH2
>2-BL3
>2-NH2T
5-A210
6-A211
BL
>1-BJ1
>1-BN1
>2-BL3(M-J)
>2-BN3
>3-BJ2
>3-BL2(M-J)
5-A110
6-GJ3
A
100-BN5
110-BL5
115-GJ1T
210-BJ5
211-BJ6
522-MT1(#12W)
523-MT2
900-NH1
930-NH1T
>960-BN1
>960-NHG2
962-NHG1
>964-BN2
>964-NHR1
A
100-ZC1
110-ZC5
>115-ZC2(#12)
210-ZD1
>211-ZD2(#12)
ZC
1-A100
>2-A115(#12)
>2-ZC6(#12)
5-A110
6-ZC2(#12)
ZD
1-A210
2-A211(#12)
LWA
2-LWB2(#12W)
A
>115-GRD (#12)
>211-GRD (#12)
522-X522(#12W)
523-X523
900-UC4
930-LD4
>960-C960
>960-UT4
962-LD5
>964-C964
>964-LD9
C
811-LD11
812-LD12
813-LD13
814-LD14
815-LD15
816-LD16
960-A960
964-A964
G
>6-LD6
>6-LD10
>6-UT2
BN
>1-A960
>1-BL1
>2-A964
>2-BN3(M-J)
>3-BL2
>3-BN2(M-J)
5-A100
6-GJ1
GJ
1-BN6
3-BL6
1T-AD1
2T-AD2
3T-A115
MT
1-A522(#12W)
2-A523
3-GRD (G)
LWB
2-LWA2(#12W)
LT
1-X521
2-X522(#12W)
X
521-LT1
>522-LT2(#12W)
X
>522-A522(#12W)
523-A523
LB
13-LF2(#18)
14-LE2(#18)
15-LC1(#18)
16-GRD (#18G)
LC
1-LB15(#18)
2-XL2(#18)
NH
1-A900
2-BJ1
1T-A930
2T-BJ2
G1-A962
>G2-A960
>G2-NHR2
R1-A964
R2-NHG2
GRD
-MT3(G)
LD
1-UC4
4-A930
5-A962
6-G6
8-UC2
9-A964
10-G6
11-C811
12-C812
13-C813
14-C814
15-C815
16-C816
LE
1-XL1(#18)
2-LB14(#18)
LF
1-XL1(#18)
2-LB13(#18)
UC
>1-UT1(#8)
>1-- 1(#8)
2-LD8
>3-UT3(#8)
>3-+ 1(#8)
>4-A900
>4-LD1
UL
1-UT1(#8)
2-XL2
3-UT3(#8)
4-XL1
UT
>1-UC1(#8)
>1-UL1(#8)
2-G6
>3-UC3(#8)
>3-UL3(#8)
4-A960
XL
>1-LE1(#18)
>1-LF1(#18)
>1-UL4
>2-LC2(#18)
>2-UL2
GRD
> -A115(#12)
> -A211(#12)
> -LB16(#18G)
- 1
-UC1(#8)
+ 1
-UC3(#8)

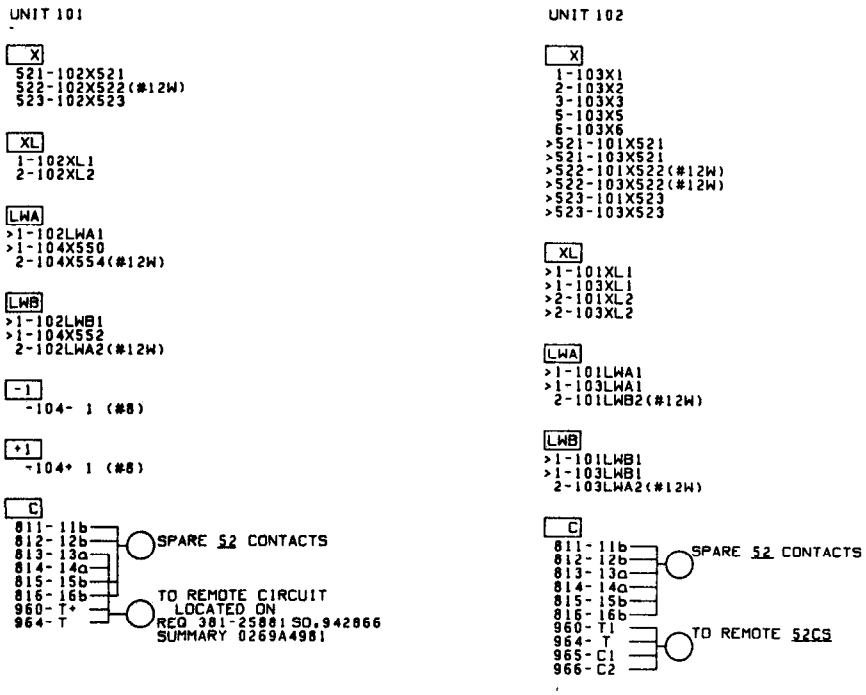


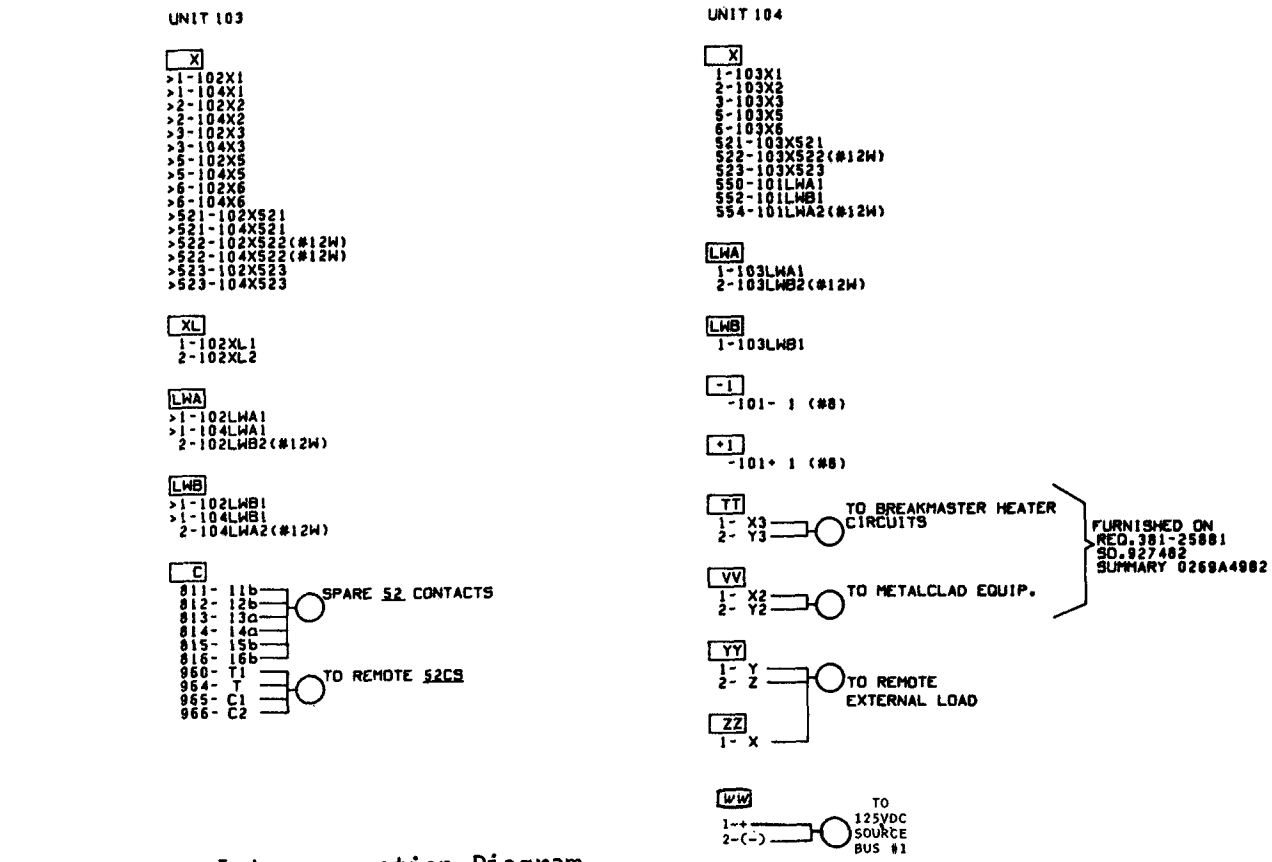
Figure 7 - Typical

Interconnection Diagrams

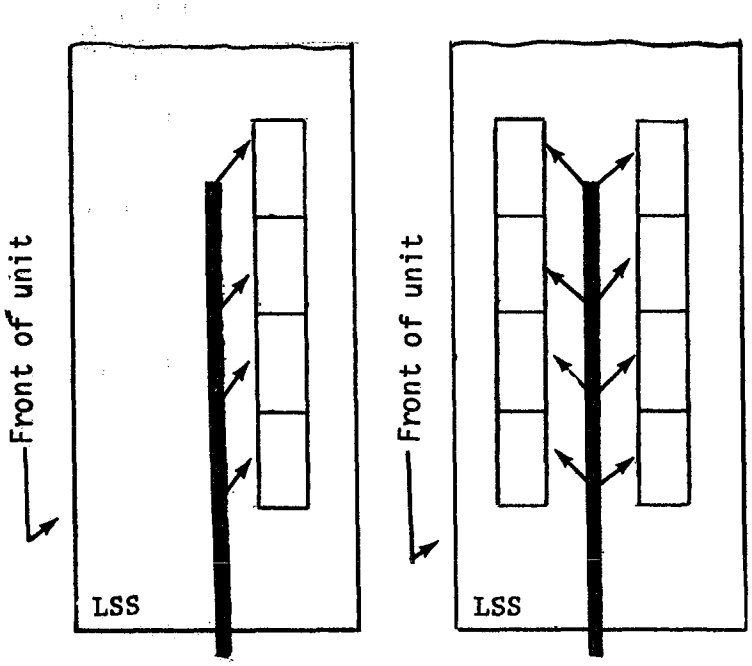
Figure 7 illustrates a typical interconnection diagram. This diagram lists the internal wiring connections between various units in the lineup under consideration. The diagram also lists those terminal points where the customer is to connect his external control and power wiring to the metal-clad switchgear lineup.

Conventions used in the preparations of this document are identical to those used in the preparation of the connection diagrams.

The terminal boards designated for customer external control and power connections are always mounted on the lower part of the left side sheet (LSS). These terminal boards are judiciously mounted to aid the customer in connecting his external control wires. Figures 8 and 9 show recommended orientation of external control wire trunking.

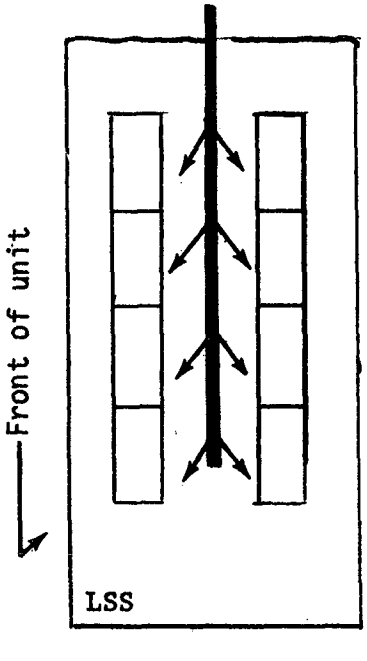


Interconnection Diagram



Customer control wires entering unit from below

Figure 8



Customer control wires entering unit from above

Figure 9

DIAGRAM ABBREVIATIONS

AC	ALTERNATING CURRENT
ACB	AIR CIRCUIT BREAKER
AIL	AMBER INDICATING LAMP
AM	AMMETER
AMP	AMPERE
ANN	ANNUNCIATOR
ARM	ARMATURE
AS	AMMETER SWITCH
AUTO TR	AUTO-TRANSFORMER
AUX	AUXILIARY
BASW	BELL ALARM SWITCH
BAT	BATTERY
BAT CHG	BATTERY CHARGER
BB	BENCHBOARD
BC	BACK CONNECTED
BD	BOARD
BE	BREAKER END
BKR	BREAKER
BT	BUS TIE
BV	BACK VIEW
CAP	CAPACITOR; CAPACITY
CAT	CATALOG
CC	CLOSING COIL
CIL	CLEAR INDICATING LAMP
CKT	CIRCUIT
CO	CUT OUT
COMPT	COMPARTMENT
CONN	CONNECT
CONT	CONNECT: CONNECTION
CPT	CONTROL POWER TRANS-FORMER
CS	CONTROL SWITCH
CT	CURRENT TRANSFORMER
CY	CYCLE
DC	DIRECT CURRENT
DD	DISCONNECTING DEVICE
DEV	DEVICE
DIAG	DIAGRAM
DIFF	DIFFERENTIAL
DIR	DIRECTION
DISC	DISCONNECT
DISCH	DISCHARGE
DM	DEMAND METER
DO	DRAWOUT
DPDT	DOUBLE POLE DOUBLE THROW
DPST	DOUBLE POLE SINGLE THROW
DS	DISCONNECTING SWITCH
DSCT	DOUBLE SECONDARY CURRENT TRANSFORMER
DWG	DRAWING
DX	DUPLEX
ELEM	ELEMENTARY
EMER	EMERGENCY
ENCL	ENCLOSURE
EQ	EQUALIZER
EQUIP	EQUIPMENT
EXC	EXCITATION; EXCITER
EXIST	EXISTING
FB	FUSE BLOCK
FC	FRONT CONNECTION
FDR	FEEDER
FLD	FIELD
FLEX	FELXIBLE
FM	FREQUENCY METER
FU	FUSE
FUT	FUTURE
FV	FRONT VIEW
FWD	FORWARD
GD	GROUND DETECTOR
GEN	GENERATOR
GIL	GREEN INDICATING LAMP
GRD	GROUND
GOV	GOVERNOR
HC	HOLDING COIL
HP	HORSE POWER
HR	HAND RESET
HTR	HEATER
IMP	IMPEDOR
IND	INDUCTION DISK UNIT
INC	INCOMING
INST	INSTANTANEOUS; INSTRUMENT
KV	KILOVOLT
KVA	KILOVOLT-AMPERE
KVAH	KILOVOLT-AMPERE-HOUR
KVAHM	KILOVOLT-AMPERE-HOUR-METER
KVAM	KILOVOLT-AMPERE-METER
KVAR	KILOVAR; REACTIVE KILOVOLT-AMPERE
KVAR HR	KILOVAR-HOUR
KWHM	KILOWATT-HOURMETER
L	LAMP; LOWERING
LA	LIGHTNING ARRESTER
LC	LATCH CHECKING SWITCH
LH	LEFT HAND
LIR	LOAD INDICATING RESISTOR
LRC	LOAD RATIO CONTROL
LS	LIMIT SWITCH
LSS	LEFT SIDE SHEET
LT	LIGHT
LTG	LIGHTING
MAM	MILLIAMETER
MAN	MANUAL
MAN OP	MANUALLY OPERATED
MC	METAL-CLAD
MECH	MECHANICAL; MECHANISM
MFR	MANUFACTURE
MG	MOTOR GENERATOR
MISC	MISCELLANEOUS
MOT	MOTOR
MTD	MOUNTED
MV	MILLIVOLT
MF	MICROFARAD
NC	NORMALLY CLOSED
NEG	NEGATIVE
NEUT	NEUTRAL
NO	NORMALLY OPEN; NUMBER
NOR	NORMAL
NP	NAMEPLATE
OC	OVERCURRENT
OCB	OIL CIRCUIT BREAKER
OPR	OPERATE
PB	PUSH BUTTON
PB STA	PUSH-BOTTON STATION
PCB	POWER CIRCUIT BREAKER
PE	PANEL END
PF	POWER FACTOR
PFD	PREFERRED
PFM	POWER FACTOR METER
PH	PHASE
PM	POLARITY MARK
PNEU	PNEUMATIC
PNL	PANEL
POS	POSITION; POSITIVE
POT	POTENTIAL
PRI	PRIMARY
PT	POTENTIAL TRANSFORMER
PU	PICK UP
PWR	POWER
RB	RELAY BLOCK
RC	RESTRAINING COIL
RCD	REVERSE CURRENT DEVICE
RE	RECEPTACLE
REAC	REACTOR
REC	RECORDING
RECL	RECLOSING
RECT	RECTIFIER
REG	REGULATOR
REQ	REQUISITION
RES	RESISTANCE; RESISTOR
REV	REVERSE; REVISE
RH	RIGHT HAND
RHEO	RHEOSTAT
RIL	RED INDICATING LAMP
RSS	RIGHT SIDE SHEET
RTD	RESISTANCE TEMPERATURE DETECTOR
SEC	SECONDARY
SECT	SECTION
SEQ	SEQUENCE
SER	SERIES
SH	SHEET; SHUNT
SI	SEAL-IN DEVICE
SOL	SOLENOID
SP	SPARE
SPDT	SINGLE POLE DOUBLE THROW
SPST	SINGLE POLE SINGLE THROW
STA	STATION; STATIONARY
STAB	STABILIZER
STD	STANDARD, STATIC TIME DELAY
SUB	SUBSTATION
SUM	SUMMARY
SUPV	SUPERVISORY
SW	SWITCH
SWBD	SWITCHBOARD
SWG BKT	SWINGING BRACKET
SWGR	SWITCHGEAR
SYM	SYMBOL
SYN	SYNCHRONISM-IZING; SYNCHRONOUS; SYNCHROSCOPE
SYN CONV	SYNCHRONOUS CONVERTER
SS	SYNCHRONIZING SWITCH
TB	TERMINAL BOARD
TC	TRIP COIL
TD	TESTING DEVICE; TIME DELAY
TDC	TIME DELAY CLOSING
TDO	TIME DELAY OPENING
TEL	TELEPHONE
TLM	TELEMETER
TEMP	TEMPERATURE
TM	TEMPERATURE METER
TPST	TRIPLE POLE SINGLE THROW
TRANS	TRANSFORMER
UVD	UNDERVOLTAGE DEVICE
V	VOLT
VA	VOLT-AMPERE
VAC	VACUUM
VAR	VOLTAGE ADJUSTING RHEOSTAT; REACTIVE VOLT-AMPERE
VARHR	VAR-HOUR
VARM	VARMETER
VM	VOLTMETER
VS	VOLTMETER SWITCH
VT	VACUUM TUBE
W	WATT; WIRE
WHDM	WATTHOUR DEMAND METER
WHM	WATTHOUR METER
WIL	WHITE INDICATING LAMP
WM	WATTMETER
X	AUXILIARY RELAY
YIL	YELLOW INDICATION LAMP

AMERICAN STANDARD DEVICE FUNCTION NUMBERS

1. Master Element
2. Time-delay Starting or Closing, Relay
3. Checking or Interlocking Relay
4. Master Contactor
5. Stopping Device
6. Starting Circuit Breaker
7. Anode Circuit Breaker
8. Control Power Disconnecting Device
9. Reversing Device
10. Unit Sequence Switch
11. Reserved for future application
12. Over-Speed Device
13. Synchronous-Speed Device
14. Under-Speed Device
15. Speed or Frequency Matching Device
16. Reserved for future application
17. Shunting or Discharge Switch
18. Accelerating or Decelerating Device
19. Starting - to-Running Transition Contactor
20. Valve
21. Distance Relay
22. Equalizer Circuit Breaker
23. Temperature Control Device
24. Reserved for future application
25. Synchronizing or Synchronous-Check Device
26. Apparatus Thermal Device
27. Undervoltage Relay
28. Flame Detector
29. Isolating Contactor
30. Annunciator Relay
31. Separate Excitation Device
32. Directional Power Relay
33. Position Switch
34. Master Sequence Device
35. Brush-Operating or Slip-Ring-Short-Circuiting Device
36. Polarity or Polarzing Voltage Device
37. Undercurrent or Underpower Relay
38. Bearing Protective Device
39. Mechanical Condition Monitor
40. Field Relay
41. Field Circuit Breaker
42. Running Circuit Breaker
43. Manual Transfer or Selector Device
44. Unit Sequence Starting Relay
45. Atmospheric Conditon Monitor
46. Reverse-Phase or Phase-Balance Current Relay
47. Phase-Sequence Voltage Relay
48. Incomplete Sequence Relay
49. Machine or Transformer Thermal Relay
50. Instantaneous Overcurrent Or Rate-of Rise Relay
51. AC Time Overcurrent Relay
52. AC Circuit Breaker
53. Exciter on DC Generator Relay
54. Reserved for future application (Use 72HS)
55. Power Factor Relay
56. Field Application Relay
57. Short-Circuiting or Grounding Device
58. Rectification Failure Relay
59. Overvoltage Relay
60. Voltage or Current Balance Relay (Use 60C when 60 V is also present.)
61. Reserved for future application
62. Time-Delay Stopping, or Opening, Relay
63. Liquid or Gas Pressure of Vacuum Relay
64. Ground Protective Relay
65. Governor
66. Notching or Jogging Device
67. AC Directional Overcurrent Vacuum Relay
68. Blocking Relay
69. Permissive Control Device
70. Rheostat
71. Liquid or Gas-Level Relay
72. DC Circuit Breaker
73. Lead-Resistor Contactor
74. Alarm Relay
75. Position Changing Mechanism
76. DC Overcurrent Relay
77. Pulse Transmitter
78. Phase Angle Measuring or Out-of-Step Protective Relay
79. AC Reclosing Relay
80. Liquid or Gas Flow Relay
81. Frequency Relay
82. DC Reclosing Relay
83. Automatic Selective Control or Transfer Relay
84. Operating Mechanism
85. Carrier or Pilot-Wire Receiver Relay
86. Locking-Out Relay
87. Differential Protecting Relay
88. Auxiliary Motor or Motor Generator
89. Line Switch
90. Regulating Device
91. Voltage Directional Relay
92. Voltage and Power Directional Relay
93. Field Changing Contactor
94. Tripping or Trip-Free Relay

95.
96.
97.
98.
99.

Used only for specific applications on individual installation where none of the assigned numbered functions from 1 to 94 are suitable.

NOTE: Alternate names such as relay, contactor, circuit breaker switch or device, may be used for any function where applicable.

NOTE: Suffix letters are used with device function numbers for various purposes; for instance, suffix N is generally used if the device is connected in the secondary neutral of current transformers, and suffixes X, Y and Z are used to denote separate auxiliary devices. Where more than one device is used for similar functions a numerical suffix is used to differentiate e.g., 52-1, 52-2 etc.

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